Speaker Information

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Talk Information

Study on Cortical Information Processing

Cortical information processing emerges from interactions among local circuits, large-scale networks, and global brain states. Our studies have shown how top-down input shapes perception (1), sleep-related activity supports memory consolidation (2), and amygdalo-cortical interactions enhance perceptual memory (3). By developing large-scale imaging approaches, we revealed cortical network architectures at single-cell resolution (4) and their reorganization during unconsciousness (5). Together, these studies will provide a multi-scale view of cortical function in perception, memory, and disease.

Research Overview (for collaboration discussion)

[Research Interests] Perception, neocortex, single-cell computation, circuit and network dynamics

[Current Research Projects] Circuit mechanisms underlying somatosensory perception, memory consolidation, and large-scale cortical network dynamics in mice and marmosets

[Techniques] We use multidisciplinary approaches, including two-photon microscopy, fiber photometry and fiberoptic imaging, wide-field and mesoscale brain imaging, optogenetic and chemogenetic manipulation, and electrophysiological recordings in awake and anesthetized animals.

[Aspects or techniques of interest for future collaboration] I am particularly interested in collaborations on computational analysis of large-scale network dynamics, development of new probes and imaging technologies using mice and marmosets.

[Short Bio]

[1] Manita et al., Neuron 2015; [2] Miyamoto et al., Science 2016; [3] Saito et al., Neuron 2025; [4] Ota et al., Neuron 2021; [5] Kiyooka et al., Cell Reports 2026